Complex Systems

Answer key

Conventions used throughout

• **Forces:** F_g = gravity (mg), F_N = normal force, F_f = kinetic friction ($F_f = \mu_k F_N$), F_A and F_{app} = applied force, T_1 and T_2 = tension in each string.

• **Method:** first decide which way the system moves. Then treat all the blocks as one system: tension is inside the system and cancels, so a = F_{net} ÷ (total mass). Then look at one block by itself to find a tension.

• **Signs (as in 2.4):** each block's + direction is the way it moves along its own surface or string. Forces are written as sizes, with + or − in the equation for their direction.

• **Final answers:** the scalar with its sign (a = +0.39 m/s^2), then the vector with [positive x], where +x is the direction of motion chosen for the system. Each unknown is isolated as a formula before the numbers go in.

• **On a slope:** the part of a force along the slope uses the angle it makes with the slope. Gravity along the slope is mg sin θ and into the slope is mg cos θ (θ = slope angle).

• g = 9.81 m/s^2. Every step keeps full precision; only final answers are rounded.

QUESTION 1

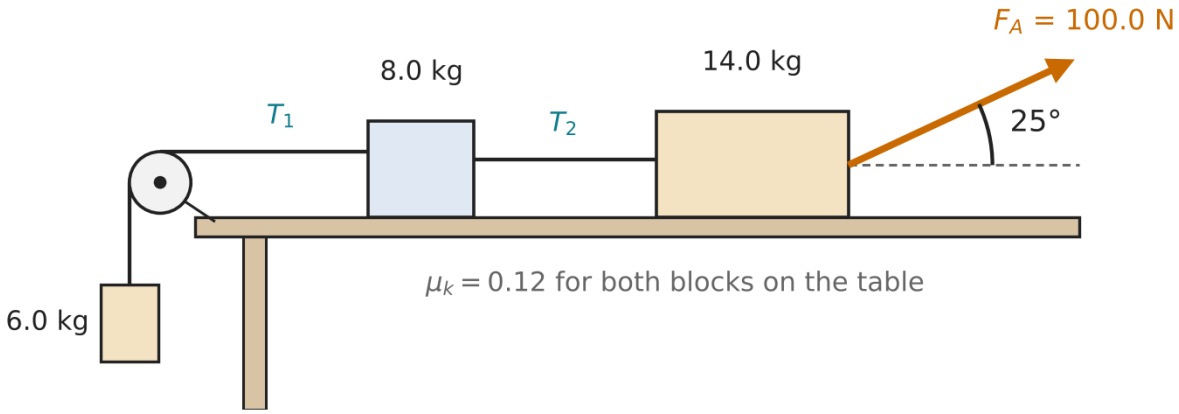
Three blocks, a pulley and an angled pull

1. A string and pulley system is set up as shown below. The string is massless and inextensible and the pulley frictionless. The coefficient of kinetic friction between the 8.0 kg and 14.0 kg blocks with the table is 0.12. Calculate:

   a) the acceleration of the 8.0 kg block [0.39 m/s^2]

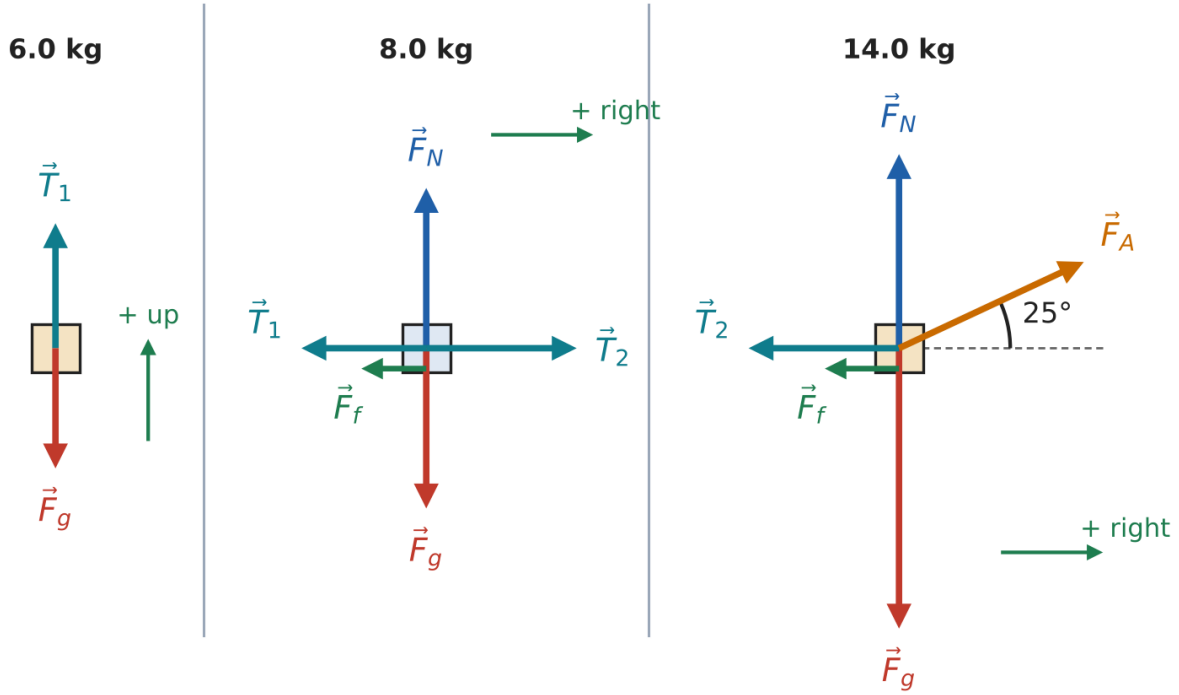
   b) the tension in the string between the 8.0 kg and 14.0 kg block. [74 N]

Picture the situation



F_A pulls the two table blocks to the right; the hanging 6.0 kg block pulls them to the left.

FBDs of each block (+ for each block is the way it moves)



(a) the acceleration of the 8.0 kg block [0.39 m/s^2]

Step 1 – Split F_A into components

The x part is next to the 25° angle (cosine); the y part is opposite it (sine):

$$F_{Ax} = (100.0)\cos 25° = 90.63 \text{ N}$$

$$F_{Ay} = (100.0)\sin 25° = 42.26 \text{ N}$$

Step 2 – Which way does the system move? (two cases)

F_{Ax} pulls the system one way and the 6.0 kg block's weight pulls it the other. Friction always opposes the motion, so it can only slow the system, never reverse it:

$$F_{g6} = (6.0)(9.81) = 58.86 \text{ N} \quad < \quad F_{Ax} = 90.63 \text{ N}$$

Case 1 (+x: table blocks right, 6.0 kg up) has the bigger pull along +x, so it gives a positive a (Step 4). **Case 2** (+x: 6.0 kg down) would need F_{g6} to beat F_{Ax} plus friction; it gives a = (58.86 − 90.63 − 9.42 − 11.41) ÷ 28.0 = -1.88 m/s^2, a negative value that friction can't produce, so Case 2 can't happen. Use Case 1.

Step 3 – Normal forces and friction

Neither table block moves up or down. F_{Ay} lifts part of the 14.0 kg block's weight, so its normal force is smaller:

$$F_{N14} = F_{g14} - F_{Ay} = (14.0)(9.81) - 42.26$$

$$F_{N14} = 137.34 - 42.26 = 95.08 \text{ N}$$

$$F_{N8} = F_{g8} = (8.0)(9.81) = 78.48 \text{ N}$$

Formula: $F_f = \mu_k F_N$

$$F_{f14} = (0.12)(95.08) = 11.41 \text{ N} \qquad F_{f8} = (0.12)(78.48) = 9.418 \text{ N}$$

Step 4 – Net force and acceleration of the system

Forward (right): F_{Ax}. Backward: the 6.0 kg block's weight and both friction forces. Tension is inside the system and cancels:

Formula: $F_{net} = (m_6 + m_8 + m_{14})\,a$

Put in the forces: $F_{Ax} - F_{g6} - F_{f8} - F_{f14} = (m_6 + m_8 + m_{14})\,a$

Divide: $$a = \frac{F\cos\theta - m_6 g - \mu_k m_8 g - \mu_k (m_{14} g - F\sin\theta)}{m_6 + m_8 + m_{14}}$$

$$a = \frac{90.63 - 58.86 - 9.418 - 11.41}{6.0 + 8.0 + 14.0} = \frac{10.94}{28.0} = +0.3908 \text{ m/s}^2$$

Every block has this acceleration, including the 8.0 kg block.

(b) the tension in the string between the 8.0 kg and 14.0 kg block. [74 N]

Look at the 14.0 kg block by itself (+ right). F_{Ax} pulls it forward; T_2 and friction pull it back:

Formula: $F_{net} = ma$

14.0 kg block: $F_{Ax} - T_2 - F_{f14} = m_{14}a$

Add T_2 and subtract $m_{14}a$ on both sides: $T_2 = F_{Ax} - F_{f14} - m_{14}a$

In letters: $T_2 = F\cos\theta - \mu_k(m_{14} g - F\sin\theta) - m_{14}a$

$T_2 = 90.63 - 11.41 - (14.0)(0.3908) = 73.75$ N

FINAL ANSWER

(a) a = +0.39 m/s^2; $\vec{a}$ = 0.39 m/s^2 [positive x], the same for all three blocks

(b) T_2 = 74 N

Check: $T_1 = F_{g6} + m_6 a = 58.86 + (6.0)(0.3908) = 61.21$ N, and the 8.0 kg block gives $T_2 - T_1 - F_{f8} = 73.75 - 61.21 - 9.418 = 3.127$ N = (8.0)(0.3908).

QUESTION 2

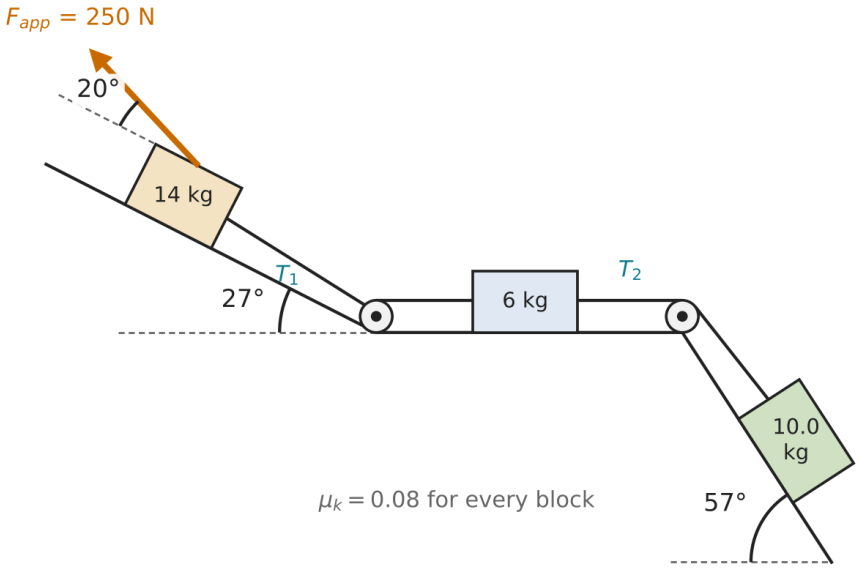
Two slopes and a flat section

2) A string and pulley system is set up as shown below. The string is massless and inextensible and the pulley frictionless. The coefficient of kinetic friction between the blocks and the surface is 0.08. Calculate:

a) the acceleration of the 6.0 kg block. [2.6 m/s^2]

b) the tension in the string between the 14.0 kg and 6.0 kg blocks. [133 N]

Picture the situation



F_{app} pulls the 14 kg block up its slope at 20° above the slope surface. T_1 joins the 14 kg and 6 kg blocks; T_2 joins the 6 kg and 10.0 kg blocks.

(a) the acceleration of the 6.0 kg block. [2.6 m/s^2]

Step 1 – Split F_{app} along and perpendicular to the slope

F_{app} makes 20° with the slope. The part along the slope is next to the 20° angle (cosine). The part perpendicular to the slope (pulling the block away from it) is opposite the angle (sine):

$$F_{app\,\parallel} = (250)\cos 20° = 234.92 \text{ N}$$

$$F_{app\,\perp} = (250)\sin 20° = 85.51 \text{ N}$$

Step 2 – Gravity along and into each slope

$$\text{14 kg:}\quad F_{g\parallel 14} = (14)(9.81)\sin 27° = 62.35\ \text{N}$$

$$F_{g\perp 14} = (14)(9.81)\cos 27° = 122.371\ \text{N}$$

$$\text{10.0 kg:}\quad F_{g\parallel 10} = (10.0)(9.81)\sin 57° = 82.27\ \text{N}$$

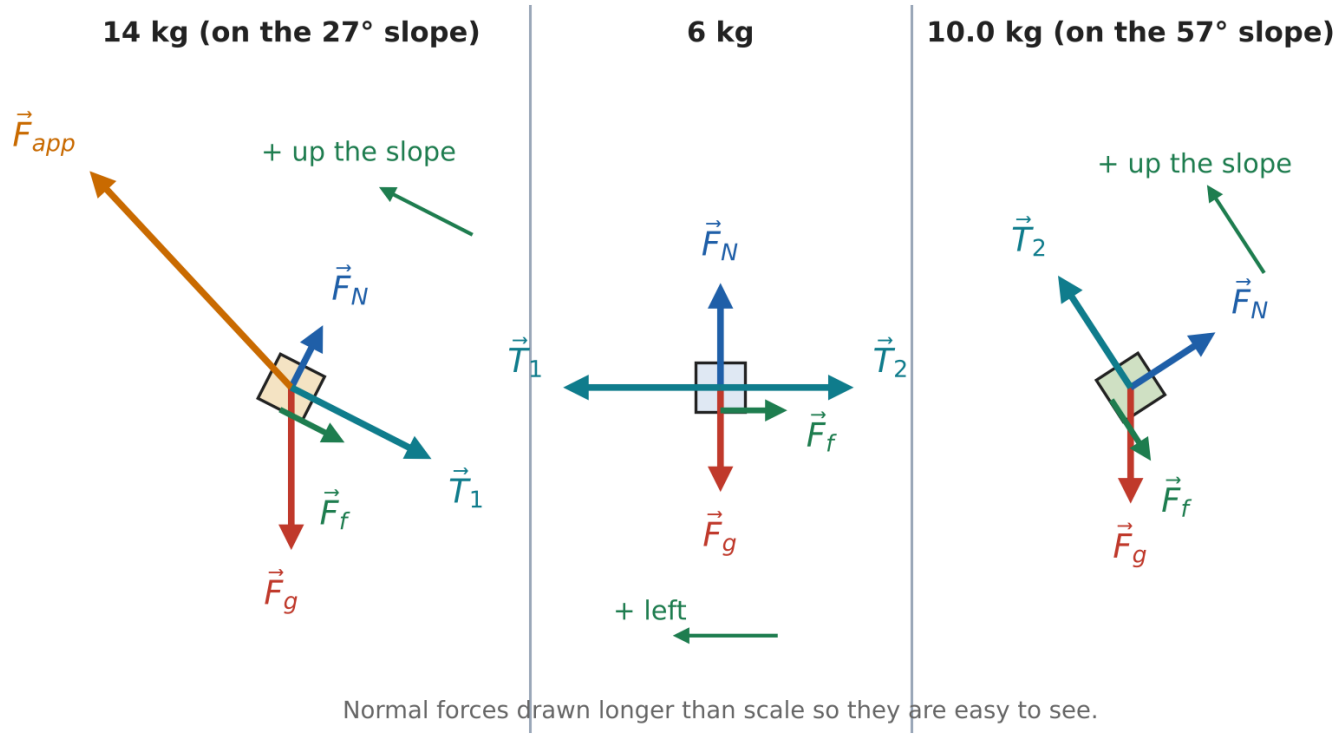
$$F_{g\perp 10} = (10.0)(9.81)\cos 57° = 53.43\ \text{N}$$

Step 3 – Which way does the system move? (two cases)

$F_{app\parallel}$ pulls the 14 kg block up its slope. The two gravity parts along the slopes pull the other way. **Case 1** (+x: 14 kg up its slope, 6 kg left, 10.0 kg up its slope) has the bigger pull along +x. **Case 2** (+x the other way) would need the gravity parts to beat $F_{app\parallel}$, which they don't, and friction can't make up the difference, so Case 2 gives a negative a and can't happen. Use Case 1:

$$F_{app\parallel} = 234.92\ \text{N} \quad > \quad 62.35 + 82.27 = 144.62\ \text{N}$$

FBDs (+ for each block is the way it moves)



Normal forces drawn longer than scale so they are easy to see.

Step 4 – Normal forces and friction

Each normal force balances everything pushing the block into its surface. For the 14 kg block, $F_{app\perp}$ pulls it away from the slope, so its normal force is smaller:

$$F_{N14} = F_{g\perp 14} - F_{app\perp} = 122.371 - 85.505 = 36.866 \text{ N}$$

$$F_{N6} = (6.0)(9.81) = 58.86 \text{ N} \qquad F_{N10} = F_{g\perp 10} = 53.43 \text{ N}$$

Formula: $F_f = \mu_k F_N$

$$F_{f14} = (0.08)(36.866) = 2.949 \text{ N} \qquad F_{f6} = (0.08)(58.86) = 4.709 \text{ N}$$

$$F_{f10} = (0.08)(53.43) = 4.274 \text{ N}$$

Step 5 – Net force and acceleration of the system

Forward: $F_{app\parallel}$. Backward: both gravity parts along the slopes and all three friction forces:

Formula: $F_{net} = (m_{14} + m_6 + m_{10})\,a$

Put in the forces:

$$F_{app\parallel} - F_{g\parallel 14} - F_{g\parallel 10} - F_{f14} - F_{f6} - F_{f10} = (m_{14} + m_6 + m_{10})\,a$$

Divide: $a = \dfrac{F_{app\parallel} - F_{g\parallel 14} - F_{g\parallel 10} - F_{f14} - F_{f6} - F_{f10}}{m_{14} + m_6 + m_{10}}$

$$a = \frac{234.92 - 62.35 - 82.27 - 2.949 - 4.709 - 4.274}{14 + 6.0 + 10.0}$$

$$a = \frac{78.37}{30.0} = +2.6122 \text{ m/s}^2$$

(b) the tension in the string between the 14.0 kg and 6.0 kg blocks. [133 N]

Look at the 14 kg block by itself (+ up its slope). $F_{app\parallel}$ pulls it up; gravity along the slope, friction and T_1 pull it down the slope:

14 kg block: $F_{app\parallel} - F_{g\parallel 14} - F_{f14} - T_1 = m_{14}a$

Add T_1 and subtract $m_{14}a$ on both sides:

$$T_1 = F_{app\parallel} - F_{g\parallel 14} - F_{f14} - m_{14}a$$

$$T_1 = 234.92 - 62.35 - 2.949 - (14)(2.6122) = 133.05 \text{ N}$$

FINAL ANSWER

(a) a = +2.6 m/s^2; $\vec{a}$ = 2.6 m/s^2 [positive x], the same for all three blocks

(b) T_1 = 1.3 × 10^2 N

Worksheet answer: (b) 133 N
Why the answers differ: The physics is the same; only the rounding differs. The unrounded tension is 133.1 N, so the worksheet's 133 N is that value to 3 figures. The given data (14 kg, 6.0 kg, 250 N) have 2 significant figures and μ = 0.08 has only 1, so 3 figures is more precision than the data allow. Rounded to 2 figures (like part (a), 2.6 m/s^2) it is 1.3 × 10^2 N.

Check: $T_2 = m_{10}a + F_{g\parallel 10} + F_{f10}$ = 112.67 N, and the 6 kg block gives $T_1 - T_2 - F_{f6}$ = 133.05 − 112.67 − 4.709 = 15.67 N = (6.0)(2.6122).